Assignment - 1

1. Explain challenges & applications of Cloud computing.

→ Cloud computing is the technology of storing and accessing data & cloud computing services over the internet. It provides on-demand services such as servers, networking, databases & storage.

- The challenges of cloud computing :

i) Internet Connectivity.

- Cloud services depend on an internet connection. Poor or unavailable connectivity can affect & performance.

ii) Vendor Lock-in.

- Users may become dependent on a particular cloud services. provider, users make it difficult for providers to move services or data to another provider.

iii) Limited Control

- Since the infrastructure is managed by the cloud provider, users may have limited control over the underlying resources.

iv) Security

- Storing & accessing data through cloud infrastructure creates security concerns.

v) Performance Dependency

- In a public cloud, performance can depend on speed of the internet connection to the cloud provida.

vi) Data Control

- In public cloud computing, the client may have limited control over data.

- The applications of cloud computing :

i) Data Storage

- Cloud computing allows users to store data on remote servers.

users can access their stored data through the internet.

iv) Data Backup & Recovery
- Cloud computing can be used for backing up & restoring data.
- Backup & restore are specifically listed as pros of cloud computing.
- Storage visualization is also used for Backup & Recovery purposes.

iii) Software Access
- SaaS allows users to access applications through the Internet using a web browser.
- The cloud provider hosts & manages the applications. Egs are eApps, Salesforce, Dropbox, Slack, etc.

iv) Application Development & Testing
- PaaS provides a platform for programmers to develop, test, run & manage applications. It supports different languages & frameworks.

2. Define Cloud computing & state its desired features.

→ Cloud computing is defined as storing & accessing data & computing services over the Internet. It provides the on-demand availability of computer services such as:
- servers.
- networking
- databases.

- The main goal of cloud computing is to provide access to data centers & remote servers to many users.

- The features of cloud computing:

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i) On-Demand Availability

Cloud computing provides computing services, whenever they are required. Users can access resources such as servers, storage & networking through the Internet.
This makes cloud resources convenient & flexible.

ii) Backup & Restore

Cloud computing provides facilities for backing up data. Data can also be restored when required. This helps protect important files from data loss.

iii) Improved Collaboration

Cloud services support collaboration among users. Multiple users can access shared cloud resources & services. This users can access shared cloud resources and services. This makes it easy for users and organization to work together.

iv) Excellent Accessibility

Cloud services can be accessed over the internet. Users can access data & services remotely. This makes cloud computing highly accessible.

v) Low Maintenance

Cloud computing can reduce the maintenance cost for users. In public cloud computing, cloud services provider maintains the infrastructure themselves.

vi) Mobility

Cloud services can be accessed remotely. Users are not necessary restricted to a particular physical location. This provides mobility to users.

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vii) Large storage capacity
- The cloud provides virtually limitless storage space.

viii) Enhanced Data Security
- It offers encryption & secure access to protect sensitive information. Security may include authentication, access control, backup & other protective measures.

ix) Pay per use model
- Users can pay according to the resources or services consumed. This reduces the need for initial investments in hardware & software.

3. Explain various layers of cloud computing.

- The cloud computing is commonly explained using 3 major services layers :
i) Infrastructure as a Service (IaaS).
ii) Platform as a Service (PaaS)
iii) Software as a Service (SaaS)

i) Infrastructure as a Service (IaaS).
- IaaS is also known as hardware as a Service. It provides computing infrastructure over the internet. The resources provided are :
- virtual machines, services networks, Processing power, virtualized hardware resources.
- In IaaS, the cloud service provider owns the physical hardware & also manages it. The user rents or uses virtualized computing resources through the Internet.
- The user can configure OS, applications, storage & networks

settings according to requirements.

- Characteristics
- Resource are available as a service
- Services are highly scalable.
- Resources are dynamic & flexible.
- Access is provided through a graphical user interface & APIs.

- Examples:
- DigitalOcean, Amazon web services, Rackspace.

ii) Platform as a Service. (Paas)

- Paas provides a cloud based platform for programmers & developers to develop, test, run & manage applications.
- In Paas, the cloud services provider manages the underlying hardware operating system, networking & runtime environments.
- Paas is based on virtualization technology, so resources can be scaled up or down accordingly.

- Characteristics
- It's accessible to various users through the same development application.
- It integrates with web services & databases.
- It supports multiple programming languages & frameworks.

- Examples
- Microsoft Azure, Apache Stratos, Openshift, Heroku.

iii) Software as a Service (Saas)

- Saas is also called on-demand software. In this model, software

applications are hosted by a cloud service provider & delivered to users through the internet.

- In SaaS, the service provider manages the hardware, OS, application software, updates, & maintenance. Users only need an Internet connection & web browser.

- Characteristics
- It has automatic updates
- Pay as you use
- The user has no hardware / software responsibility.
- Example
  - DropBox, Slack, Salesforce, Google Apps.

5. Write & explain various types of deployment models of Cloud computing.*

→ The cloud deployment defines how cloud services are made available. The various types of deployment models are:

i) Public Cloud
- It is open to all users, managed by CSPs, pay-per-use.
- The pros / advantages are: low cost, scalable, location independent.
- The cons / disadvantages are: less secure, limited control.
- Examples of Public cloud are: Amazon EC2, Google App Engine.

ii) Private Cloud
- Internal / corporate cloud managed by the organization or third party.
- The advantages are: High Security, Better performance,

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full control.
- The disadvantages are required skilled staff, limited scalability
- Examples are : Openstack, Eucalyptus

iii) Hybrid Cloud.
- It is a combination of public and private cloud.
- The pros / advantages are : flexible resources, faster product delivery, reduced risk.
- The cons / disadvantages are complex management, reliability depends on providers.
- Example
- Office 365, AWS, Google Suite.

iv) Community Cloud
- It is shared by multiple organization with common goals.
- The advantages are : cost-effective, collaborative, better security than public cloud
- The disadvantages are limited storage / bandwidth, not suitable for all organizations.
- Examples : Healthcare community cloud.

5. What is Virtualization & its pros in Cloud computing.

→ Virtualization is the creation of virtual version of resources such as servers, desktops, storage devices, OS or networks.
- It allows multiple users and organization to share a single physical resource efficiently.

- Pros :

i) Efficient Resource Utilization.
- maximizes hardware usage by running multiple virtual machine

iii) Cost reduction
- It eliminates the need for multiple physical servers.

iii) Scalability
- Resources can be scaled up or down easily.

iv) Flexibility
- supports different OS and apps on the same hardware.

v) Simplified management
- It is easier to control virtual machines compared to physical servers.

vi) Backup & Recovery
- Provides better disaster recovery solutions.

6. Explain types of virtualization in brief.

→ The types of virtualization
i) Hardware
- It allows multiple OS to run on the same hardware. The main job of hypervisor is to control & monitoring processor, memory.
- It is mainly used for servers platforms.

ii) OS
- The VMM is installed on OS host instead of hardware. It is used for testing applications on diff. OS platform.

iii) Server
- The VMM is installed on server systems. It divides one physical server into multiple virtual servers. It helps in load balancing & resource allocation.

iv) Storage
- It groups multiple physical storage devices into one virtual storage. It is implemented using software applications. It is mainly used for backup & recovery purposes.

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